

VOICE 1: IDLE

Thu Jan 1, 12:46 am



Bash Training

ACM SIG Cyber

```
:q  
# ls  
test.c  
# cc test.c  
# ls  
a.out test.c  
# a.out  
hello world!#
```

Agenda

- Definition
- Importance
- Key Features
- Benefits
- Labs
- Closure

What is Bash?

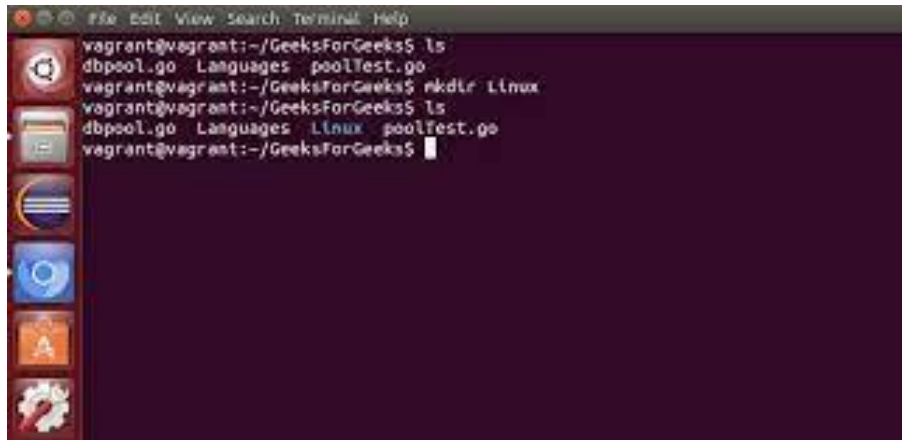
- Bash (Bourne-Again SHell)
 - Extremely used and powerful shell and scripting language
- Command-Line Interface (CLI)
 - Allowing to interact with the OS through commands
- Usually the default shell
- Scripting Language



Shell vs Scripts

Shell:

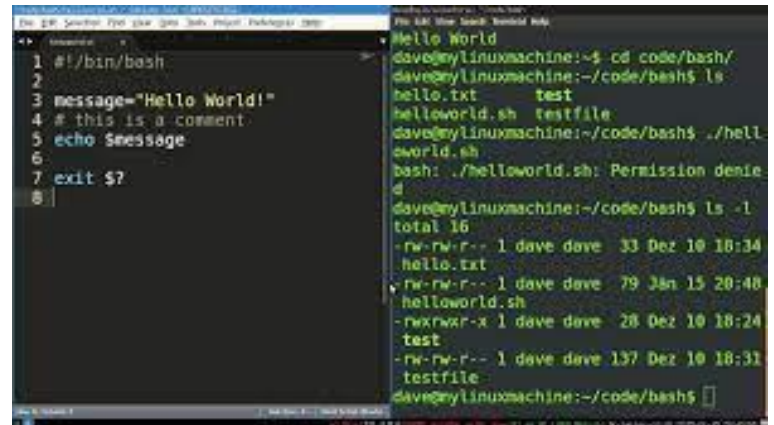
- Interactive command-line interface for immediate command execution
- Real-time System management
- Quick tasks



```
File Edit View Search Terminal Help
vagrant@vagrant:~/GeeksForGeeks$ ls
dbpool.go Languages poolTest.go
vagrant@vagrant:~/GeeksForGeeks$ mkdir Linux
vagrant@vagrant:~/GeeksForGeeks$ ls
dbpool.go Languages Linux poolTest.go
vagrant@vagrant:~/GeeksForGeeks$
```

Scripts:

- Text files containing sequences of commands, mainly for automation
- Automate tasks for repetitive, complex operations



```
File Edit View Search Terminal Help
1 #!/bin/bash
2
3 message="Hello World!"
4 # this is a comment
5 echo $message
6
7 exit $?
8
dave@mylinuxmachine:~$ cd code/bash/
dave@mylinuxmachine:~/code/bash$ ls
hello.txt test
dave@mylinuxmachine:~/code/bash$ ./helloworld.sh testfile
bash: ./helloworld.sh: Permission denied
dave@mylinuxmachine:~/code/bash$ ls -l
total 16
-rw-rw-r-- 1 dave dave 33 Dec 10 18:34 hello.txt
-rw-rw-r-- 1 dave dave 79 Jan 15 20:48 helloworld.sh
-rwxrwxr-x 1 dave dave 28 Dec 10 18:24 test
-rw-rw-r-- 1 dave dave 137 Dec 10 18:31 testfile
dave@mylinuxmachine:~/code/bash$
```

Bash in Cybersecurity

Sooooooo many uses in security..

Bash plays a huge role in cybersecurity for various tasks, enhancing efficiency and security

- Automation: Repetitive tasks -> reduce human errors
 - Automating log analysis, security monitoring, etc
- Network Monitoring: Scripts can create custom monitoring tools
 - Monitoring system logs for suspicious activities
- Incident Response: Bash facilitates quick responses to security incidents
 - Isolate compromised system, collect forensic data, and initiate countermeasures
- Penetration Testing: Automate tasks
 - Vulnerability scanning, reporting findings

The ability to use Bash effectively is an **essential** skill in cybersecurity!

Fundamental Bash Commands

Before we go to scripting, we need to familiarize ourselves with commands

`echo`: printing to the terminal ex: `echo "Hello World"` -> "Hello World"

`cd`: changing directories (folders) ex: `cd path/to/directory` changes the working directory to the specified path

`ls`: list directory contents (files in directories) ex: `ls` or `ls -l` lists files/directories in long format

`mkdir`: creating directories ex: `mkdir new_directory`

`touch`: create new empty file ex: `touch new_file.txt`

`chmod`: changing file permissions ex: `chmod 755 my_script.sh` (755 = read,write,execute)

`rm`: removing files/directories ex: `rm file.txt`

Basic Structure of Bash Scripts

To write effective Bash scripts, it's essential to understand their structure and key components.

Every script starts with

- (shebang line) `#!/bin/bash` - specifying the interpreter

Comments = `#`

Variables: Essential for storing data/results

- User-defined variables (`my_variable="Hello"`) or pre-defined system vars (`$HOME`)
- Referenced using `$` ex (`echo $my_variable`)

Functions: functions group commands to reuse

- ex: `function_name() { commands }`

Control Structures: `if`, `elif`, `else`

- Handle different conditions

Example Script: Hello World

Feel free to follow along

Flexibility of Bash

Bash scripts are extremely flexible and portable making it extremely common for administrators and developers

Runs on Unix-like systems (Linux/macOS)

- Tool like Git Bash, Cygwin, etc helps execute bash scripts on Windows

Script Reusability

- Custom functions and libraries

Open-Source Ecosystem

- Allows developers to access community-created tools

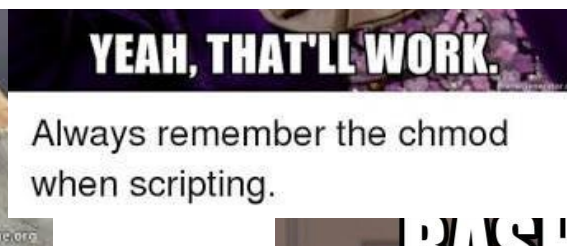
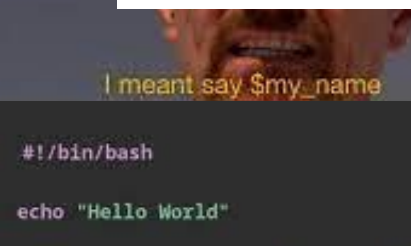
Practical Benefits of Bash

- Simple to create.
 - Straightforward to write/understand
- Can run multiple commands easily.
 - Improving efficiency
- Interactive debugging.
 - Helps identify bugs fast
- Portable.
 - Compatible
- Time-saving.
 - Automation of repetitive tasks
- Cost-effective.
 - Free and open-source, reducing software related expenses

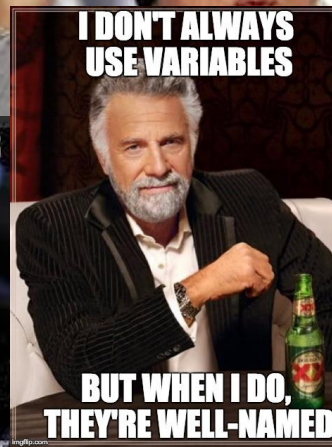


spend
5 minutes
ing a
e task

That was a lot! Questions? Comments?



spend
half an hour
automating
a simple task



Guided Lab: Activity Generator w/ User Input

Objective: Create a bash script that generates activities and invites the user to participate

1. Define Activities
2. User Interaction
3. Script Execution

Independent Lab: Create a basic Login System

Objective: Create a bash script that simulates a basic login system

For assistance:

1. Set predefined credentials
2. Script implementation
 - a. Prompt the user to enter username/pw
 - b. Compare entered credentials with pre-defined
 - c. Provide feedback based on outcome
3. Script Execution

Challenge to previous lab...

Implement Account Lockout

- Modify the login system to include an account log out mechanism
 - If a user enters incorrect credentials for a specified number of times, their account should be temporarily locked

Resources

[Bash Scripting Cheat Sheet](#)

[Bash Scripting GitHub](#)

[Linux Documentation](#) (GOAT)